

4.5V 至 18V 输入，6A 同步降压转换器

查询样品: [TPS54627](#)

特性

- **D-CAP2™** 模式支持快速瞬态响应
- 低输出纹波，支持陶瓷输出电容器
- 宽 V_{IN} 输入电压范围: **4.5V 至 18V**
- 输出电压范围: **0.76V 至 5.5V**
- 高效率集成型场效应晶体管 (FET)
针对较低占空比应用进行了优化
- **36mΩ** (高侧) 与 **28mΩ** (低侧)
- 高效率，关断时流耗少于 **10μA**
- 高初始带隙基准精度
- 可调软启动
- 预偏置软启动
- **650kHz** 开关频率 (f_{sw})
- 逐周期过流限制

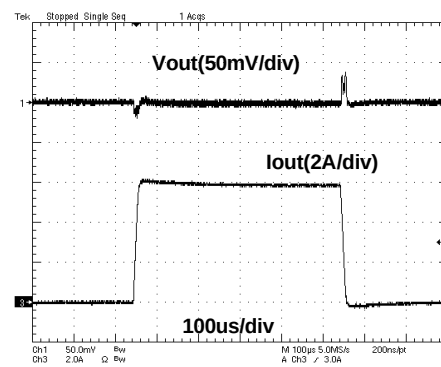
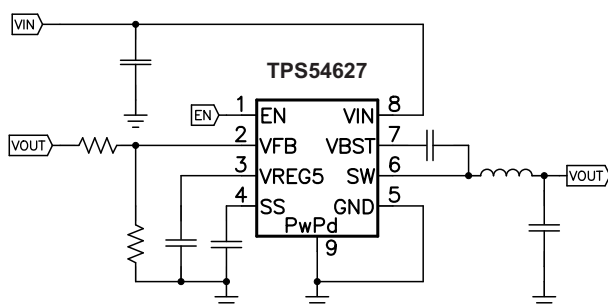
应用范围

- 低电压系统的广泛应用
 - 数字电视电源
 - 高清 **Blu-ray Disc™** 播放器
 - 网络家庭终端设备
 - 数字机顶盒 (STB)

说明

TPS54627 是一款自适应接通时间 D-CAP2™ 模式同步降压转换器。TPS54627 可帮助系统设计人员通过一个成本有效、低组件数量、低待机电流解决方案来完成多种终端设备的电源总线调节器集。TPS54627 的主控制环路使用 D-CAP2™ 模式控制，此控制方式在无需外部补偿组件的情况下提供快速瞬态响应。

TPS54627 的专有电路还使该器件可采用诸如高分子有机半导体固体电容器 (POSCAP) 或高分子聚合物电容器 (SP-CAP) 等低等效串联电阻 (ESR) 输出电容器，以及超低 ESR 陶瓷电容器。该器件的工作输入电压介于 4.5V 至 18V V_{IN} 之间。输出电压可在 0.76V 与 5.5V 之间进行设定。该器件还特有一个可调软启动时间。TPS54627 采用 8 引脚 DDA 封装，并针对在 -40°C 到 85°C 的工作温度范围内的运行而设计。



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Blu-ray Disc is a trademark of Blu-ray Disc Association.

PRODUCTION DATA information is current as of publication date. Products conform to specifications per the terms of the Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.

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These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

ORDERING INFORMATION⁽¹⁾

T _A	PACKAGE ⁽²⁾ ⁽³⁾	ORDERABLE PART NUMBER	PIN	TRANSPORT MEDIA
–40°C to 85°C	DDA	TPS54627DDA	8	Tube
		TPS54627DDAR		Tape and Reel

- (1) For the most current package and ordering information, see the Package Option Addendum at the end of this document, or see the TI web site at www.ti.com.
(2) Package drawings, thermal data, and symbolization are available at www.ti.com/packaging.
(3) All package options have Cu NIPDAU lead/ball finish.

ABSOLUTE MAXIMUM RATINGS

over operating free-air temperature range (unless otherwise noted) ⁽¹⁾

		VALUE		UNIT
		MIN	MAX	
Input voltage range	VIN, EN	–0.3	20	V
	VBST	–0.3	26	
	VBST (10 ns transient)	–0.3	28	
	VBST (vs SW)	–0.3	6.5	
	VFB, SS	–0.3	6.5	
	SW	–2	20	
	SW (10 ns transient)	–3	22	
Output voltage range	VREG5	–0.3	6.5	V
	GND	–0.3	0.3	
Voltage from GND to thermal pad, V _{diff}		–0.2	0.2	V
Electrostatic discharge	Human Body Model (HBM)		2	kV
	Charged Device Model (CDM)		500	V
Operating junction temperature, T _J		–40	150	°C
Storage temperature, T _{stg}		–55	150	

- (1) Stresses beyond those listed under *absolute maximum ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *recommended operating conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

THERMAL INFORMATION

THERMAL METRIC ⁽¹⁾		TPS54627	UNITS
		DDA (8 PINS)	
θ _{JA}	Junction-to-ambient thermal resistance	43.5	°C/W
θ _{JCtop}	Junction-to-case (top) thermal resistance	49.4	
θ _{JB}	Junction-to-board thermal resistance	25.6	
ψ _{JT}	Junction-to-top characterization parameter	7.4	
ψ _{JB}	Junction-to-board characterization parameter	25.5	
θ _{JCbot}	Junction-to-case (bottom) thermal resistance	5.2	

- (1) For more information about traditional and new thermal metrics, see the IC Package Thermal Metrics application report, [SPRA953](http://www.ti.com/lit/an/spra953).

RECOMMENDED OPERATING CONDITIONS

over operating free-air temperature range, (unless otherwise noted)

			MIN	MAX	UNIT
V_{IN}	Supply input voltage range		4.5	18	V
V_I	Input voltage range	VBST	–0.1	24	V
		VBST (10 ns transient)	–0.1	27	
		VBST(vs SW)	–0.1	6.0	
		SS	–0.1	5.7	
		EN	–0.1	18	
		VFB	–0.1	5.5	
		SW	–1.8	18	
		SW (10 ns transient)	–3	21	
		GND	–0.1	0.1	
V_O	Output voltage range	VREG5	–0.1	5.7	V
I_O	Output Current range	I_{VREG5}	0	5	mA
T_A	Operating free-air temperature		–40	85	°C
T_J	Operating junction temperature		–40	150	°C

ELECTRICAL CHARACTERISTICS

over operating free-air temperature range, $V_{IN} = 12$ V (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
SUPPLY CURRENT						
I_{VIN}	Operating - non-switching supply current	V_{IN} current, $T_A = 25^\circ\text{C}$, EN = 5 V, $V_{FB} = 0.8$ V		950	1400	μA
I_{VINSN}	Shutdown supply current	V_{IN} current, $T_A = 25^\circ\text{C}$, EN = 0 V		3	10	μA
LOGIC THRESHOLD						
V_{EN}	EN high-level input voltage	EN	1.6			V
	EN low-level input voltage	EN			0.6	V
R_{EN}	EN pin resistance to GND	$V_{EN} = 12$ V	200	400	800	k Ω
V_{FB} VOLTAGE AND DISCHARGE RESISTANCE						
V_{FBTH}	V_{FB} threshold voltage	$T_A = 25^\circ\text{C}$, $V_O = 1.05$ V, continuous mode operation	757	765	773	mV
		$T_A = -40$ to 85°C , $V_O = 1.05$ V, continuous mode operation ⁽¹⁾	751	765	779	mV
I_{VFB}	V_{FB} input current	$V_{FB} = 0.8$ V, $T_A = 25^\circ\text{C}$		0	± 0.15	μA
V_{REG5} OUTPUT						
V_{VREG5}	V_{REG5} output voltage	$T_A = 25^\circ\text{C}$, 6.0 V < V_{IN} < 18 V, $0 < I_{VREG5} < 5$ mA	5.2	5.5	5.7	V
I_{VREG5}	Output current	$V_{IN} = 6$ V, $V_{REG5} = 4.0$ V, $T_A = 25^\circ\text{C}$	20			mA
V_{OUT} DISCHARGE						
R_{DISCHG}	V_{OUT} discharge resistance	EN = 0 V, SW = 0.5 V, $T_A = 25^\circ\text{C}$		500	800	Ω
MOSFET						
$R_{DS(on)}$	High side switch resistance	25°C , $V_{BST} - SW = 5.5$ V		36		m Ω
	Low side switch resistance	25°C		28		m Ω
CURRENT LIMIT						
I_{OCL}	Current limit	L out = 1.5 μH ⁽¹⁾	6.7	7.3	8.9	A

(1) Not production tested.

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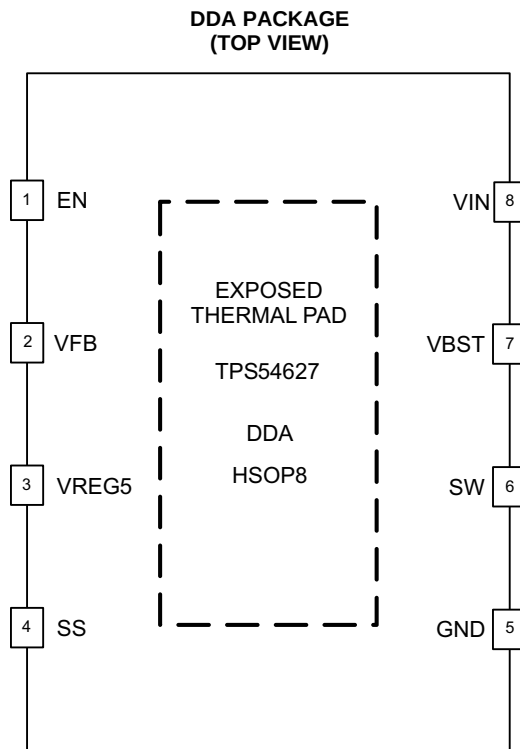
ELECTRICAL CHARACTERISTICS (continued)

 over operating free-air temperature range, $V_{IN} = 12\text{ V}$ (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
THERMAL SHUTDOWN						
T _{SDN}	Thermal shutdown threshold	Shutdown temperature ⁽²⁾	165			°C
		Hysteresis ⁽²⁾	35			
ON-TIME TIMER CONTROL						
t _{ON}	On time	V _{IN} = 12 V, V _O = 1.05 V	150			ns
t _{OFF(MIN)}	Minimum off time	T _A = 25°C, V _{FB} = 0.7 V	260 310			ns
SOFT START						
I _{SS}	SS charge current	V _{SS} = 1 V	4.2	6	7.8	μA
	SS discharge current	V _{SS} = 0.5 V	1.5	3.3		mA
HICCUP AND OVER-VOLTAGE PROTECTION						
V _{OVP}	Output OVP threshold	OVP Detect (L > H)	125%			
V _{HICCUP}	Output Hiccup threshold	Hiccup detect (H > L)	65%			
T _{HICCUPDELAY}	Output Hiccup delay	To hiccup state	250			μsec
T _{HICCUPENDELAY}	Output Hiccup Enable delay	Relative to soft-start time	x1.7			
UVLO						
UVLO	UVLO threshold	Wake up V _{REG5} voltage	3.45	3.75	4.05	V
		Hysteresis V _{REG5} voltage	0.13	0.32	0.48	

(2) Not production tested.

DEVICE INFORMATION



PIN FUNCTIONS

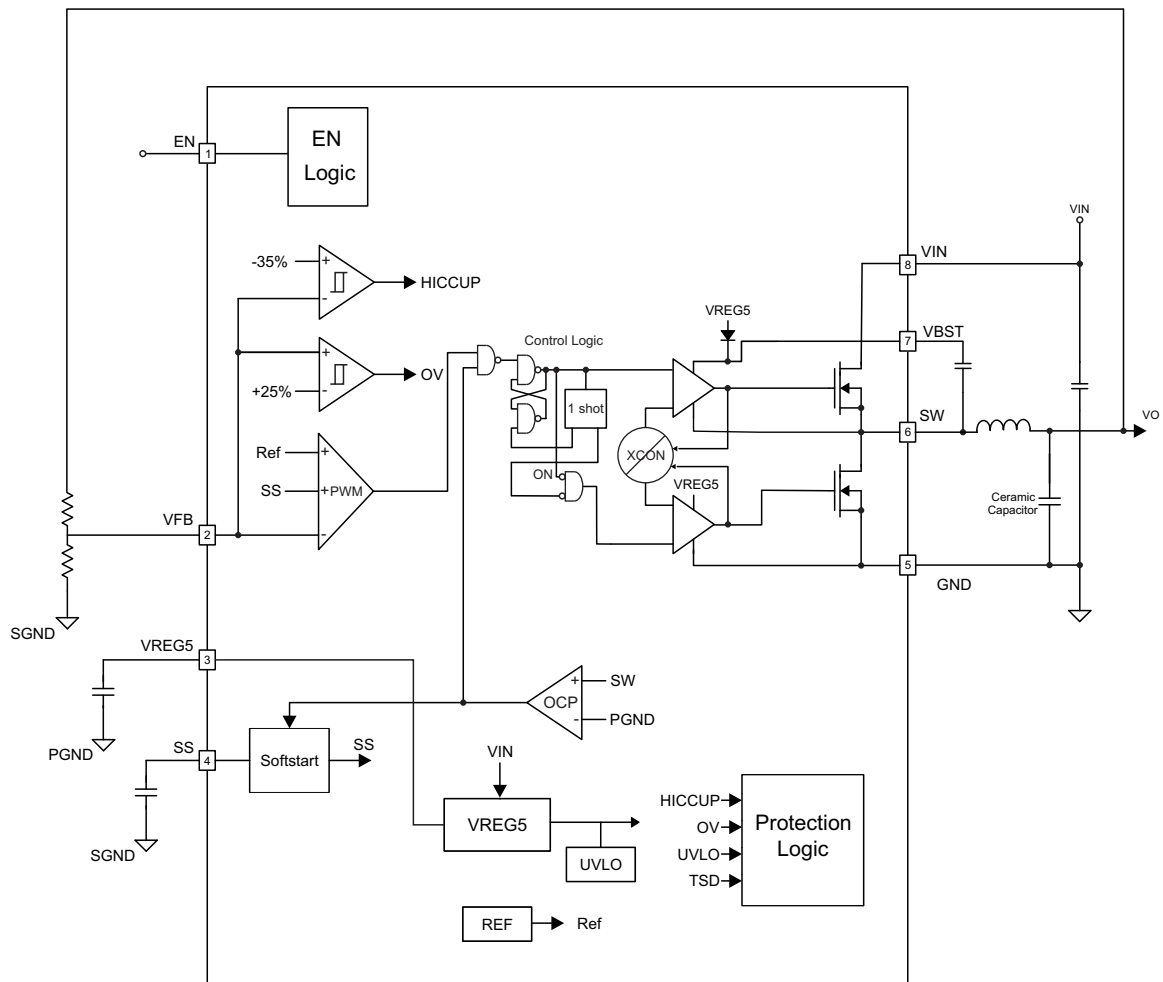
PIN		DESCRIPTION
NAME	NO.	
EN	1	Enable input control. EN is active high and must be pulled up to enable the device.
VFB	2	Converter feedback input. Connect to output voltage with feedback resistor divider.
VREG5	3	5.5 V power supply output. A capacitor (typical 1 μ F) should be connected to GND. VREG5 is not active when EN is low.
SS	4	Soft-start control. An external capacitor should be connected to GND.
GND	5	Ground pin. Power ground return for switching circuit. Connect sensitive SS and VFB returns to GND at a single point.
SW	6	Switch node connection between high-side NFET and low-side NFET.
VBST	7	Supply input for the high-side FET gate drive circuit. Connect 0.1 μ F capacitor between VBST and SW pins. An internal diode is connected between VREG5 and VBST.
VIN	8	Input voltage supply pin.
Exposed Thermal Pad	Back side	Thermal pad of the package. Must be soldered to achieve appropriate dissipation. Must be connected to GND.

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FUNCTIONAL BLOCK DIAGRAM



OVERVIEW

The TPS54627 is a 6-A synchronous step-down (buck) converter with two integrated N-channel MOSFETs. It operates using D-CAP2™ mode control. The fast transient response of D-CAP2™ control reduces the output capacitance required to meet a specific level of performance. Proprietary internal circuitry allows the use of low ESR output capacitors including ceramic and special polymer types.

DETAILED DESCRIPTION

PWM Operation

The main control loop of the TPS54627 is an adaptive on-time pulse width modulation (PWM) controller that supports a proprietary D-CAP2™ mode control. D-CAP2™ mode control combines constant on-time control with an internal compensation circuit for pseudo-fixed frequency and low external component count configuration with both low ESR and ceramic output capacitors. It is stable even with virtually no ripple at the output.

At the beginning of each cycle, the high-side MOSFET is turned on. This MOSFET is turned off after internal one shot timer expires. This one shot is set by the converter input voltage, VIN, and the output voltage, VO, to maintain a pseudo-fixed frequency over the input voltage range, hence it is called adaptive on-time control. The one-shot timer is reset and the high-side MOSFET is turned on again when the feedback voltage falls below the reference voltage. An internal ramp is added to reference voltage to simulate output ripple, eliminating the need for ESR induced output ripple from D-CAP2™ mode control.

PWM Frequency and Adaptive On-Time Control

TPS54627 uses an adaptive on-time control scheme and does not have a dedicated on board oscillator. The TPS54627 runs with a pseudo-constant frequency of 650 kHz by using the input voltage and output voltage to set the on-time one-shot timer. The on-time is inversely proportional to the input voltage and proportional to the output voltage; therefore, when the duty ratio is VOUT/VIN, the frequency is constant.

Soft Start and Pre-Biased Soft Start

The soft start function is adjustable. When the EN pin becomes high, 6μA current begins charging the capacitor which is connected from the SS pin to GND. Smooth control of the output voltage is maintained during start up. The equation for the slow start time is shown in [Equation 1](#). VFB voltage is 0.765 V and SS pin source current is 6μA.

$$t_{ss}(ms) = \frac{C6(nF) \times V_{FB} \times 1.1}{I_{SS}(\mu A)} = \frac{C6(nF) \times 0.765 \times 1.1}{6} \quad (1)$$

The TPS54627 contains a unique circuit to prevent current from being pulled from the output during startup if the output is pre-biased. When the soft-start commands a voltage higher than the pre-bias level (internal soft start becomes greater than feedback voltage VFB), the controller slowly activates synchronous rectification by starting the first low side FET gate driver pulses with a narrow on-time. It then increments that on-time on a cycle-by-cycle basis until it coincides with the time dictated by (1-D), where D is the duty cycle of the converter. This scheme prevents the initial sinking of the pre-bias output, and ensure that the out voltage (VO) starts and ramps up smoothly into regulation and the control loop is given time to transition from pre-biased start-up to normal mode operation.

Output Discharge Control

TPS54627 discharges the output when EN is low, or the controller is turned off by the UVLO protection. The internal low-side MOSFET is not turned on for the output discharge operation to avoid the possibility of causing negative voltage at the output.

Current Protection

The output overcurrent protection (OCP) is implemented using a cycle-by-cycle valley detect control circuit. The switch current is monitored by measuring the low-side FET switch voltage between the SW pin and GND. This voltage is proportional to the switch current. To improve accuracy, the voltage sensing is temperature compensated.

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During the on time of the high-side FET switch, the switch current increases at a linear rate determined by V_{IN} , V_{OUT} , the on-time and the output inductor value. During the on time of the low-side FET switch, this current decreases linearly. The average value of the switch current is the load current I_{OUT} . The TPS54627 constantly monitors the low-side FET switch voltage, which is proportional to the switch current, during the low-side on-time. If the measured voltage is above the voltage proportional to the current limit, an internal counter is incremented per each SW cycle and the converter maintains the low-side switch on until the measured voltage is below the voltage corresponding to the current limit at which time the switching cycle is terminated and a new switching cycle begins. In subsequent switching cycles, the on-time is set to a fixed value and the current is monitored in the same manner. If the over current condition exists for 7 consecutive switching cycles, the internal OCL threshold is set to a lower level, reducing the available output current. When a switching cycle occurs where the switch current is not above the lower OCL threshold, the counter is reset and the OCL limit is returned to the higher value.

There are some important considerations for this type of over-current protection. The peak current is the average load current plus one half of the peak-to-peak inductor current. The valley current is the average load current minus one half of the peak-to-peak inductor current. Since the valley current is used to detect the over-current threshold, the load current is higher than the over-current threshold. Also, when the current is being limited, the output voltage tends to fall. When the VFB voltage becomes lower than 65% of the target voltage, the UVP comparator detects it. If the under-voltage condition persists for 250 μ s, the device will shut down and re-start in hiccup mode after 7 times the SS period. When the over current condition is removed, the output voltage will return to the regulated value. This protection is non-latching.

Over Voltage Protection

TPS54627 detects over and under voltage conditions by monitoring the feedback voltage (VFB). This function is enabled after approximately 1.7 x times the soft start time. When the feedback voltage becomes higher than 125% of the target voltage, the OVP comparator output goes high and both the high-side MOSFET driver and the low-side MOSFET driver turn off. This function is non-latch operation.

UVLO Protection

Undervoltage lock out protection (UVLO) monitors the voltage of the V_{REG5} pin. When the V_{REG5} voltage is lower than UVLO threshold voltage, the TPS54627 is shut off. This protection is non-latching.

Thermal Shutdown

TPS54627 monitors the temperature of itself. If the temperature exceeds the threshold value (typically 165°C), the device is shut off. This is non-latch protection.

TYPICAL CHARACTERISTICS

$V_{IN} = 12\text{ V}$, $T_A = 25^\circ\text{C}$ (unless otherwise noted).

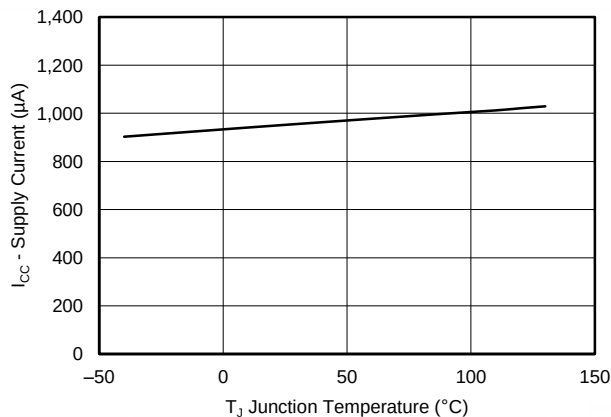


Figure 1. SUPPLY CURRENT vs JUNCTION TEMPERATURE

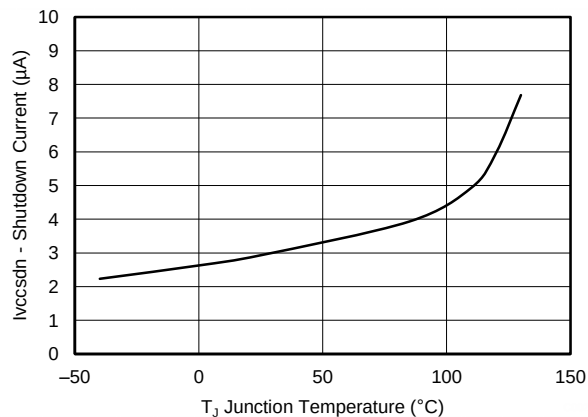


Figure 2. SHUTDOWN CURRENT vs JUNCTION TEMPERATURE

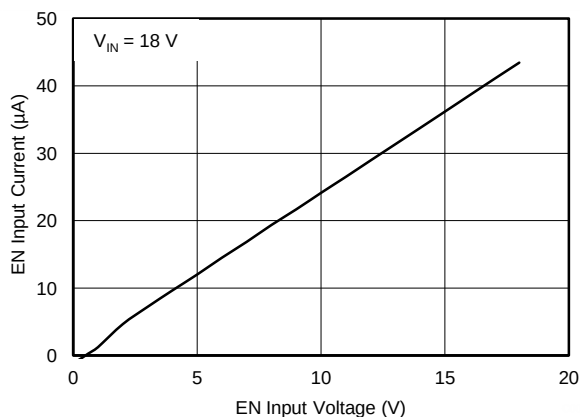


Figure 3. EN INPUT CURRENT vs EN INPUT VOLTAGE

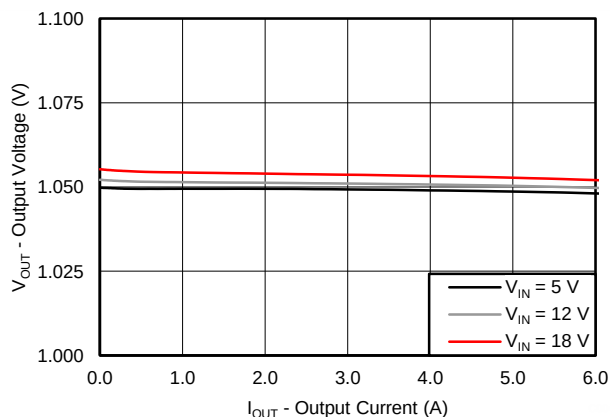


Figure 4. 1.05-V OUTPUT VOLTAGE vs OUTPUT CURRENT

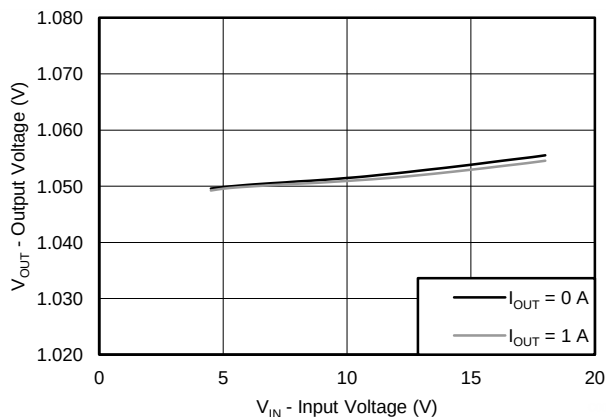


Figure 5. 1.05-V OUTPUT VOLTAGE vs INPUT VOLTAGE

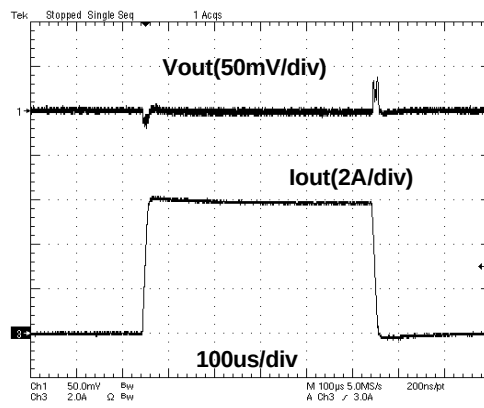


Figure 6. 1.05-V, LOAD TRANSIENT RESPONSE

TYPICAL CHARACTERISTICS (continued)

$V_{IN} = 12\text{ V}$, $T_A = 25^\circ\text{C}$ (unless otherwise noted).

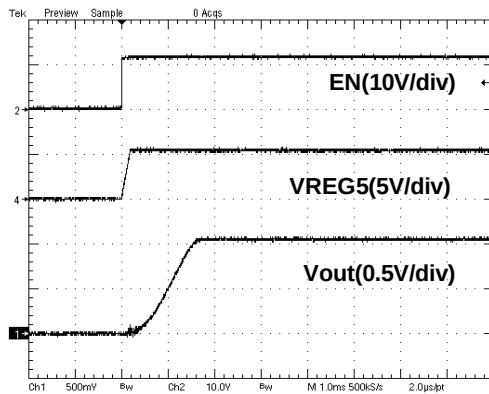


Figure 7. START-UP WAVE FORM

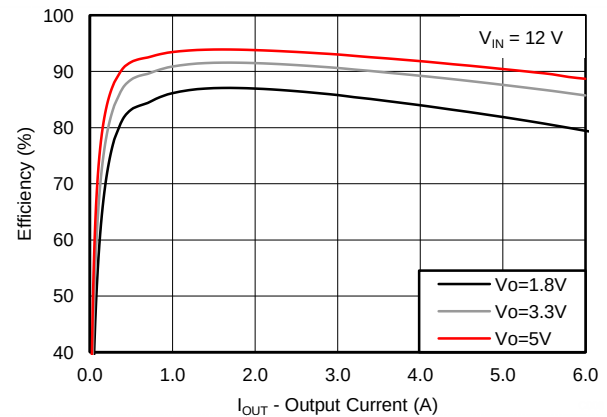


Figure 8. EFFICIENCY vs OUTPUT CURRENT

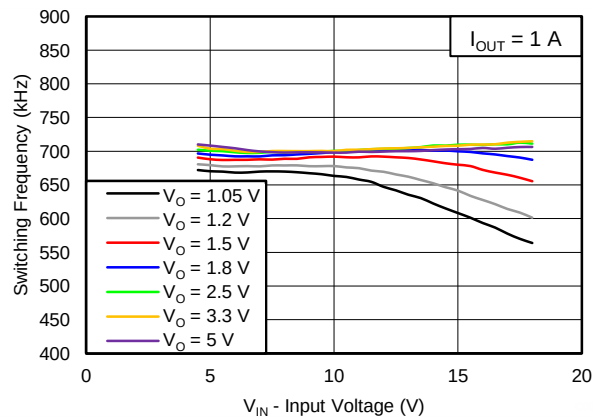


Figure 9. SWITCHING FREQUENCY vs INPUT VOLTAGE

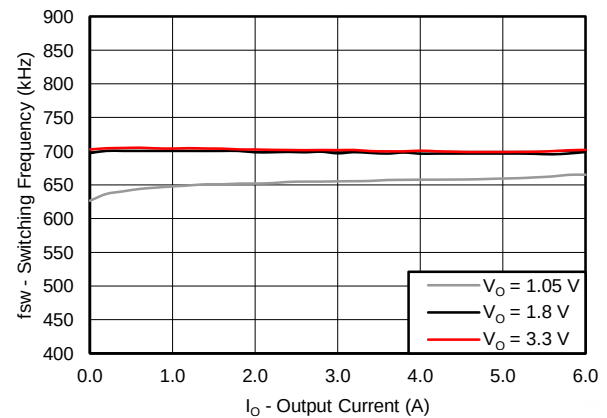


Figure 10. SWITCHING FREQUENCY vs OUTPUT CURRENT

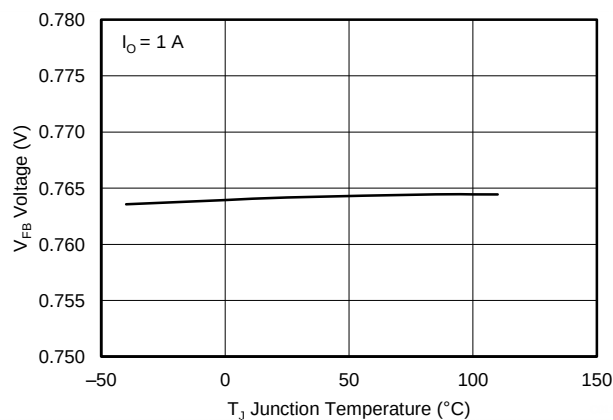


Figure 11. VFB VOLTAGE vs JUNCTION TEMPERATURE

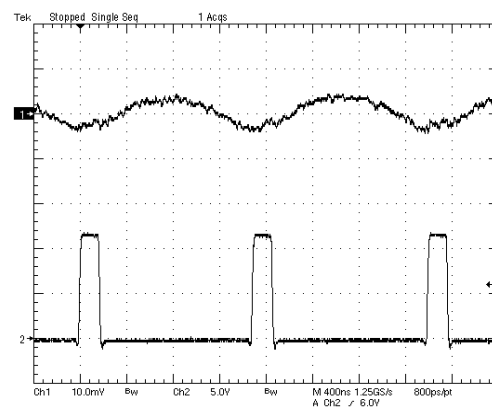


Figure 12. VOLTAGE RIPPLE AT OUTPUT ($I_O = 6\text{ A}$)

TYPICAL CHARACTERISTICS (continued)

VIN = 12 V, TA = 25°C (unless otherwise noted).

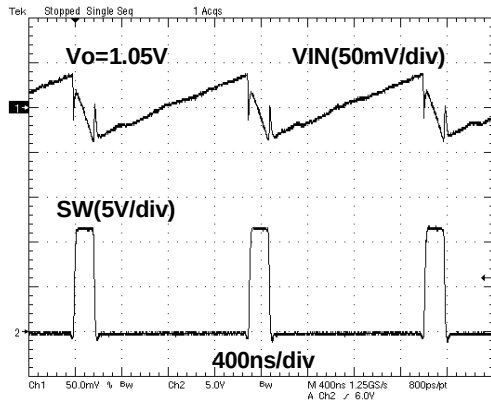


Figure 13. VOLTAGE RIPPLE AT INPUT (IO = 6 A)

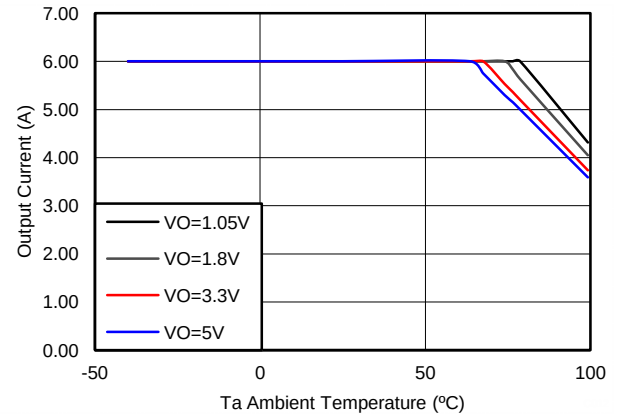


Figure 14. OUTPUT CURRENT vs AMBIENT TEMPERATURE

DESIGN GUIDE

Step-By-Step Design Procedure

To begin the design process, the user must know a few application parameters:

- Input voltage range
- Output voltage
- Output current
- Output voltage ripple
- Input voltage ripple

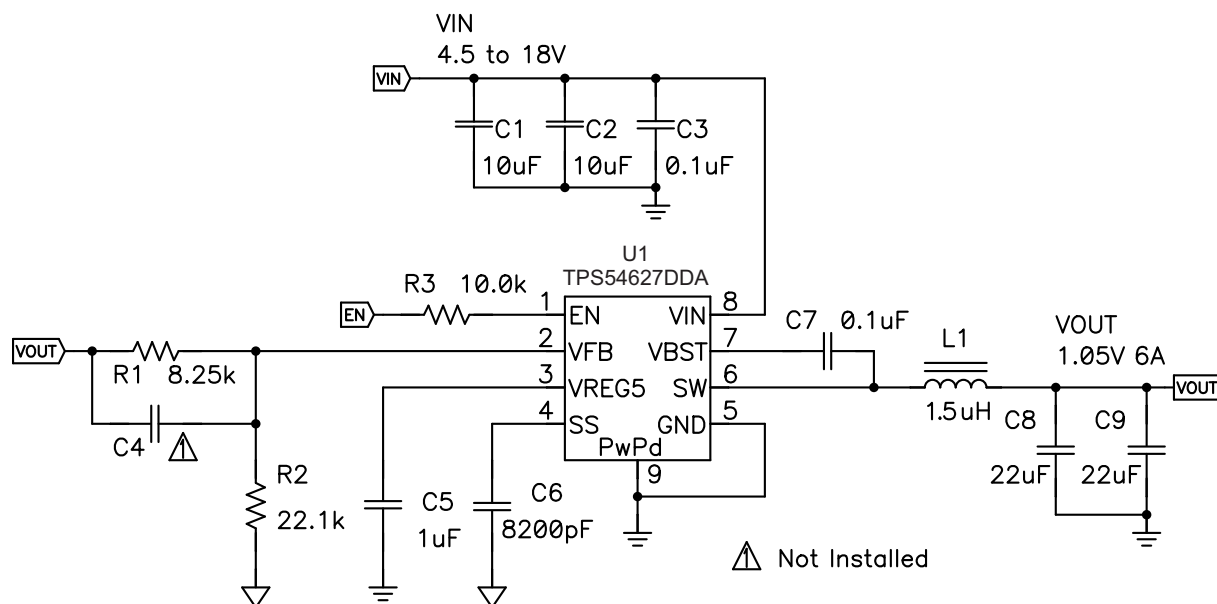


Figure 15. Shows the schematic diagram for this design example.

Output Voltage Resistors Selection

The output voltage is set with a resistor divider from the output node to the VFB pin. It is recommended to use 1% tolerance or better divider resistors. Start by using [Equation 2](#) to calculate V_{OUT} .

To improve efficiency at light loads consider using larger value resistors, high resistance is more susceptible to noise, and the voltage errors from the VFB input current are more noticeable.

$$V_{OUT} = 0.765 \times \left(1 + \frac{R1}{R2} \right) \quad (2)$$

Output Filter Selection

The output filter used with the TPS54627 is an LC circuit. This LC filter has double pole at:

$$F_P = \frac{1}{2\pi\sqrt{L_{OUT} \times C_{OUT}}} \quad (3)$$

At low frequencies, the overall loop gain is set by the output set-point resistor divider network and the internal gain of the TPS54627. The low frequency phase is 180 degrees. At the output filter pole frequency, the gain rolls off at a -40 dB per decade rate and the phase drops rapidly. D-CAP2™ introduces a high frequency zero that reduces the gain roll off to -20 dB per decade and increases the phase to 90 degrees one decade above the zero frequency. The inductor and capacitor selected for the output filter must be selected so that the double pole of [Equation 3](#) is located below the high frequency zero but close enough that the phase boost provided by the high frequency zero provides adequate phase margin for a stable circuit. To meet this requirement use the values recommended in [Table 1](#).

Table 1. Recommended Component Values

Output Voltage (V)	R1 (kΩ)	R2 (kΩ)	C4 (pF) ⁽¹⁾			L1 (μH)			C8 + C9 (μF)	
			MIN	TYP	MAX	MIN	TYP	MAX	MIN	MAX
1	6.81	22.1	5	150	220	1.0	1.5	4.7	22	68
1.05	8.25	22.1	5	150	220	1.0	1.5	4.7	22	68
1.2	12.7	22.1	5		100	1.0	1.5	4.7	22	68
1.5	21.5	22.1	5		68	1.0	1.5	4.7	22	68
1.8	30.1	22.1	5		22	1.2	1.5	4.7	22	68
2.5	49.9	22.1	5		22	1.5	2.2	4.7	22	68
3.3	73.2	22.1	2		22	1.8	2.2	4.7	22	68
5	124	22.1	2		22	2.2	3.3	4.7	22	68

(1) Optional

Since the DC gain is dependent on the output voltage, the required inductor value increases as the output voltage increases. Additional phase boost can be achieved by adding a feed forward capacitor (C4) in parallel with R1. The feed forward capacitor is most effective for output voltages at or above 1.8 V.

The inductor peak-to-peak ripple current, peak current and RMS current are calculated using Equation 4, Equation 5 and Equation 6. The inductor saturation current rating must be greater than the calculated peak current and the RMS or heating current rating must be greater than the calculated RMS current. Use 650 kHz for f_{SW} .

Use 650 kHz for f_{SW} . Make sure the chosen inductor is rated for the peak current of Equation 5 and the RMS current of Equation 6.

$$I_{IPP} = \frac{V_{OUT}}{V_{IN(max)}} \times \frac{V_{IN(max)} - V_{OUT}}{L_O \times f_{SW}} \quad (4)$$

$$I_{Ipeak} = I_O + \frac{I_{IPP}}{2} \quad (5)$$

$$I_{Lo(RMS)} = \sqrt{I_O^2 + \frac{1}{12} I_{IPP}^2} \quad (6)$$

For this design example, the calculated peak current is 6.51 A and the calculated RMS current is 6.01 A. The inductor used is a TDK SPM6530-1R5M100 with a peak current rating of 11.5 A and an RMS current rating of 11 A.

The capacitor value and ESR determines the amount of output voltage ripple. The TPS54627 is intended for use with ceramic or other low ESR capacitors. Recommended values range from 22μF to 68μF. Use Equation 7 to determine the required RMS current rating for the output capacitor.

$$I_{Co(RMS)} = \frac{V_{OUT} \times (V_{IN} - V_{OUT})}{\sqrt{12} \times V_{IN} \times L_O \times f_{SW}} \quad (7)$$

For this design two TDK C3216X5R0J226M 22μF output capacitors are used. The typical ESR is 2 mΩ each. The calculated RMS current is 0.284 A and each output capacitor is rated for 4A.

Input Capacitor Selection

The TPS54627 requires an input decoupling capacitor and a bulk capacitor is needed depending on the application. A ceramic capacitor over 10 μF is recommended for the decoupling capacitor. An additional 0.1 μF capacitor (C3) from pin 8 to ground is optional to provide additional high frequency filtering. The capacitor voltage rating needs to be greater than the maximum input voltage.

Bootstrap Capacitor Selection

A 0.1 μF. ceramic capacitor must be connected between the VBST to SW pin for proper operation. It is recommended to use a ceramic capacitor.

VREG5 Capacitor Selection

A 1- μ F. ceramic capacitor must be connected between the VREG5 to GND pin for proper operation. It is recommended to use a ceramic capacitor.

THERMAL INFORMATION

This 8-pin DDA package incorporates an exposed thermal pad that is designed to be directly to an external heatsink. The thermal pad must be soldered directly to the printed board (PCB). After soldering, the PCB can be used as a heatsink. In addition, through the use of thermal vias, the thermal pad can be attached directly to the appropriate copper plane shown in the electrical schematic for the device, or alternatively, can be attached to a special heatsink structure designed into the PCB. This design optimizes the heat transfer from the integrated circuit (IC).

For additional information on the exposed thermal pad and how to use the advantage of its heat dissipating abilities, see the Technical Brief, PowerPAD™ Thermally Enhanced Package, Texas Instruments Literature No. [SLMA002](#) and Application Brief, PowerPAD™ Made Easy, Texas Instruments Literature No. [SLMA004](#).

The exposed thermal pad dimensions for this package are shown in the following illustration.

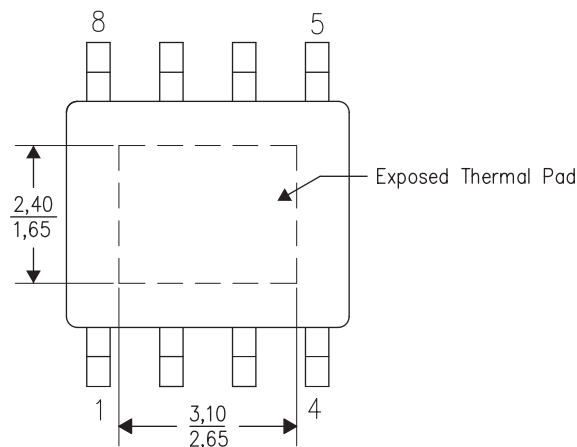
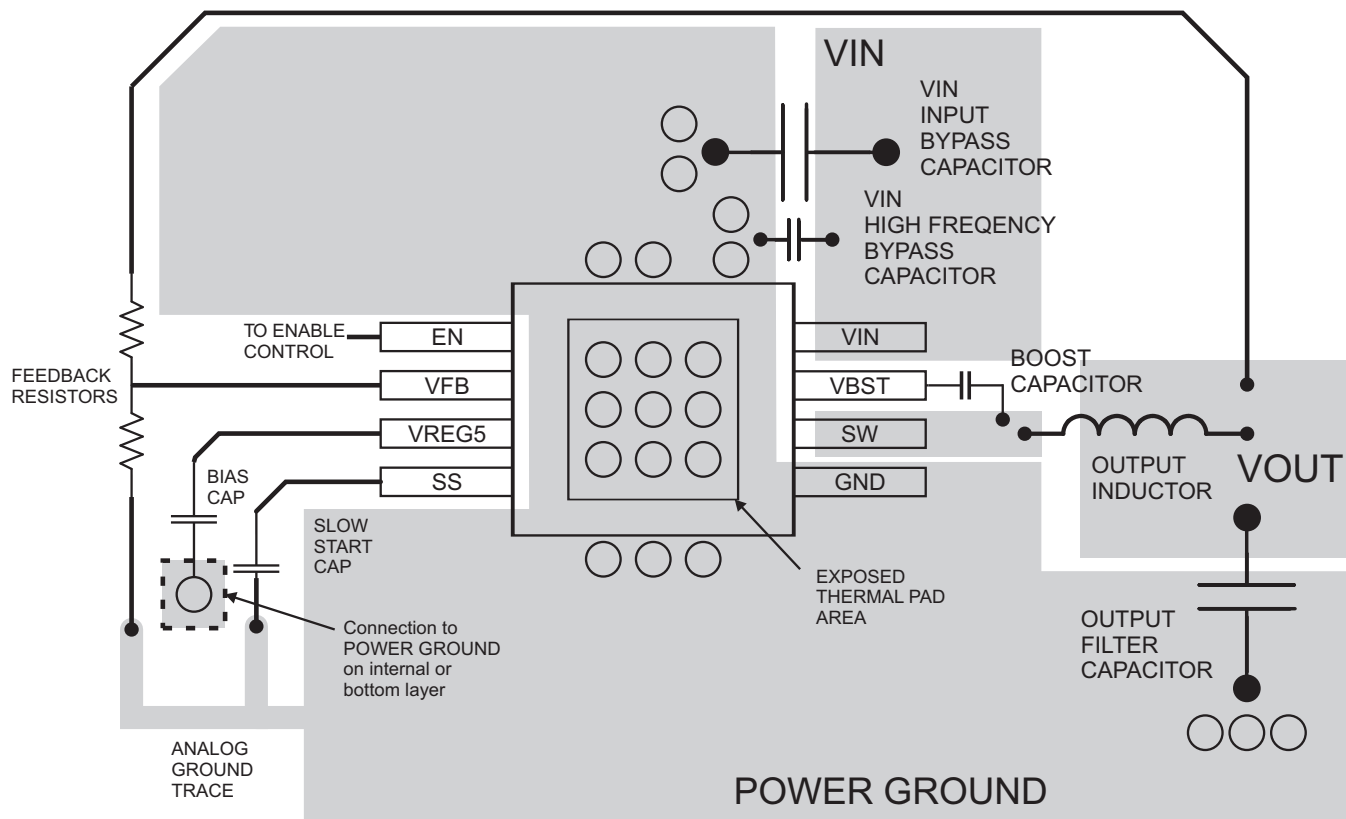


Figure 16. Thermal Pad Dimensions

LAYOUT CONSIDERATIONS

1. The TPS54627 can supply large load currents up to 6 A, so heat dissipation may be a concern. The top side area adjacent to the TPS54627 should be filled with ground as much as possible to dissipate heat.
2. The bottom side area directly below the IC should a dedicated ground area. It should be directly connected to the thermal pad of the device using vias as shown. The ground area should be as large as practical. Additional internal layers can be dedicated as ground planes and connected to the vias as well.
3. Keep the input switching current loop as small as possible.
4. Keep the SW node as physically small and short as possible to minimize parasitic capacitance and inductance and to minimize radiated emissions. Kelvin connections should be brought from the output to the feedback pin of the device.
5. Keep analog and non-switching components away from switching components.
6. Make a single point connection from the signal ground to power ground.
7. Do not allow switching current to flow under the device.
8. Keep the pattern lines for VIN and PGND broad.
9. Exposed pad of device must be connected to PGND with solder.
10. VREG5 capacitor should be placed near the device, and connected PGND.
11. Output capacitor should be connected to a broad pattern of the PGND.
12. Voltage feedback loop should be as short as possible, and preferably with ground shield.
13. Lower resistor of the voltage divider which is connected to the VFB pin should be tied to SGND.
14. Providing sufficient via is preferable for VIN, SW and PGND connection.
15. PCB pattern for VIN, SW, and PGND should be as broad as possible.
16. VIN Capacitor should be placed as near as possible to the device.



○ VIA to Ground Plane

Figure 17. PCB Layout

REVISION HISTORY

Changes from Original (April 2013) to Revision A	Page
• 已删除特性：轻负载下实现高效率的自动跳跃 Eco-mode™	1

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
TPS54627DDA	Active	Production	SO PowerPAD (DDA) 8	75 TUBE	Yes	NIPDAU SN	Level-2-260C-1 YEAR	-40 to 85	54627
TPS54627DDA.A	Active	Production	SO PowerPAD (DDA) 8	75 TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	54627
TPS54627DDAR	Active	Production	SO PowerPAD (DDA) 8	2500 LARGE T&R	Yes	NIPDAU SN	Level-2-260C-1 YEAR	-40 to 85	54627
TPS54627DDAR.A	Active	Production	SO PowerPAD (DDA) 8	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	54627
TPS54627DDARG4	Active	Production	SO PowerPAD (DDA) 8	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	54627
TPS54627DDARG4.A	Active	Production	SO PowerPAD (DDA) 8	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	54627

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

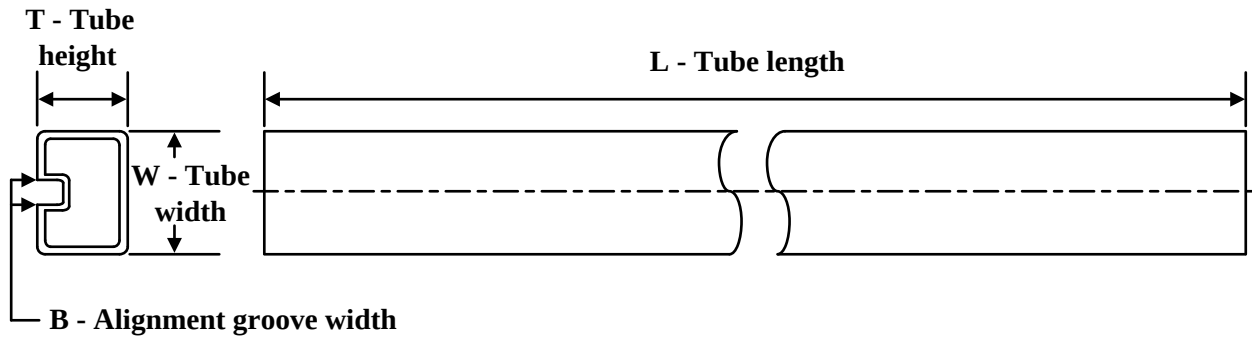
⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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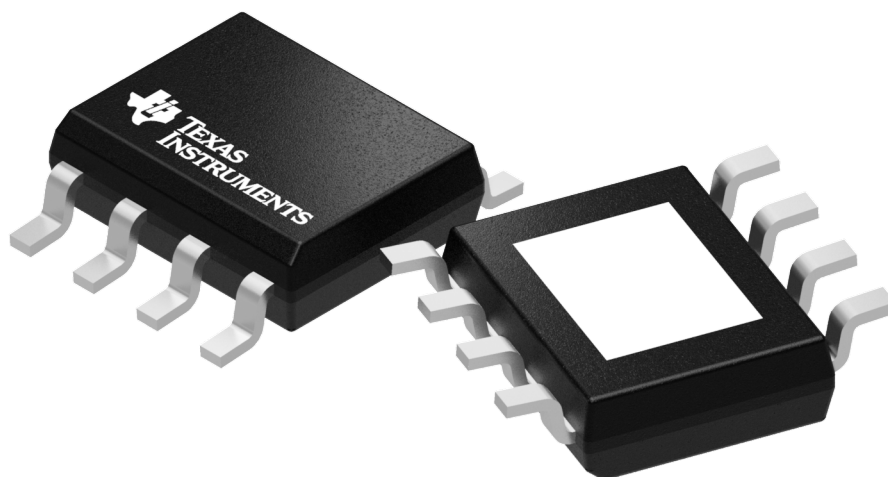
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TUBE



*All dimensions are nominal

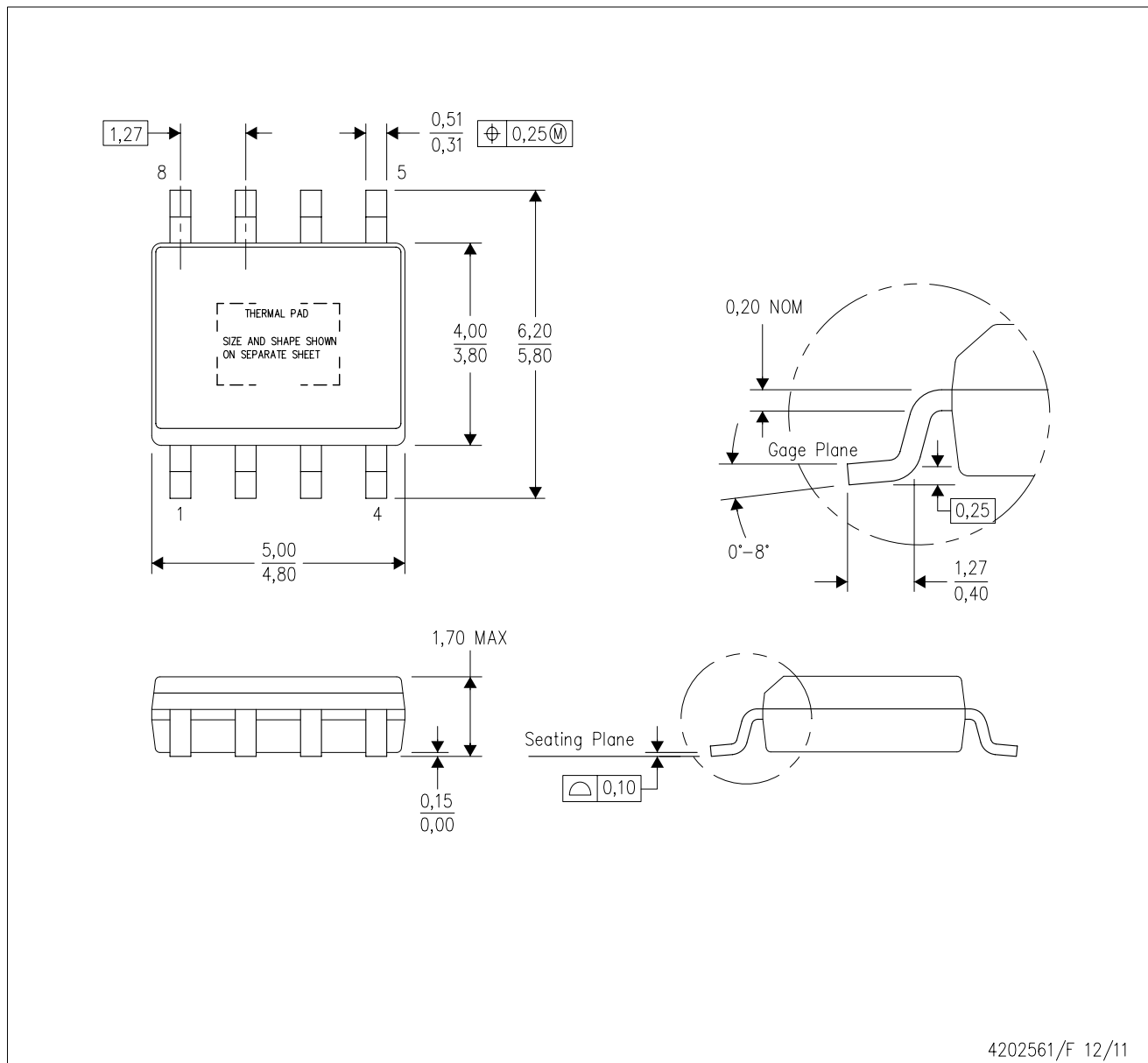
Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
TPS54627DDA	DDA	HSOIC	8	75	517	7.87	635	4.25
TPS54627DDA	DDA	HSOIC	8	75	508	7.77	2540	NA
TPS54627DDA.A	DDA	HSOIC	8	75	508	7.77	2540	NA
TPS54627DDA.A	DDA	HSOIC	8	75	517	7.87	635	4.25



Images above are just a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

DDA (R-PDSO-G8)

PowerPAD™ PLASTIC SMALL-OUTLINE



- NOTES:
- All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5-1994.
 - This drawing is subject to change without notice.
 - Body dimensions do not include mold flash or protrusion not to exceed 0,15.
 - This package is designed to be soldered to a thermal pad on the board. Refer to Technical Brief, PowerPad Thermally Enhanced Package, Texas Instruments Literature No. SLMA002 for information regarding recommended board layout. This document is available at www.ti.com <<http://www.ti.com>>.
 - See the additional figure in the Product Data Sheet for details regarding the exposed thermal pad features and dimensions.
 - This package complies to JEDEC MS-012 variation BA

PowerPAD is a trademark of Texas Instruments.

DDA (R-PDSO-G8)

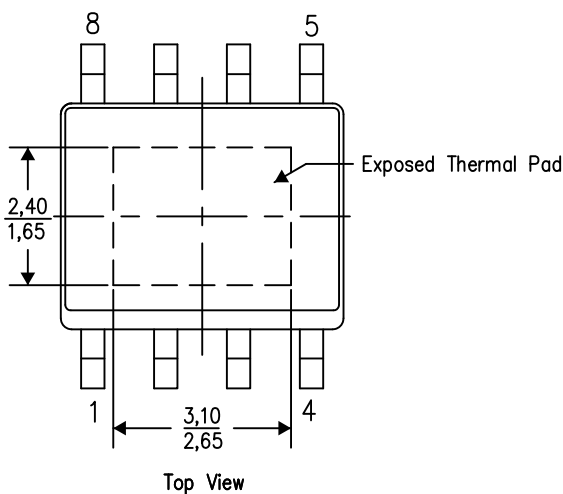
PowerPAD™ PLASTIC SMALL OUTLINE

THERMAL INFORMATION

This PowerPAD™ package incorporates an exposed thermal pad that is designed to be attached to a printed circuit board (PCB). The thermal pad must be soldered directly to the PCB. After soldering, the PCB can be used as a heatsink. In addition, through the use of thermal vias, the thermal pad can be attached directly to the appropriate copper plane shown in the electrical schematic for the device, or alternatively, can be attached to a special heatsink structure designed into the PCB. This design optimizes the heat transfer from the integrated circuit (IC).

For additional information on the PowerPAD package and how to take advantage of its heat dissipating abilities, refer to Technical Brief, PowerPAD Thermally Enhanced Package, Texas Instruments Literature No. SLMA002 and Application Brief, PowerPAD Made Easy, Texas Instruments Literature No. SLMA004. Both documents are available at www.ti.com.

The exposed thermal pad dimensions for this package are shown in the following illustration.



Exposed Thermal Pad Dimensions

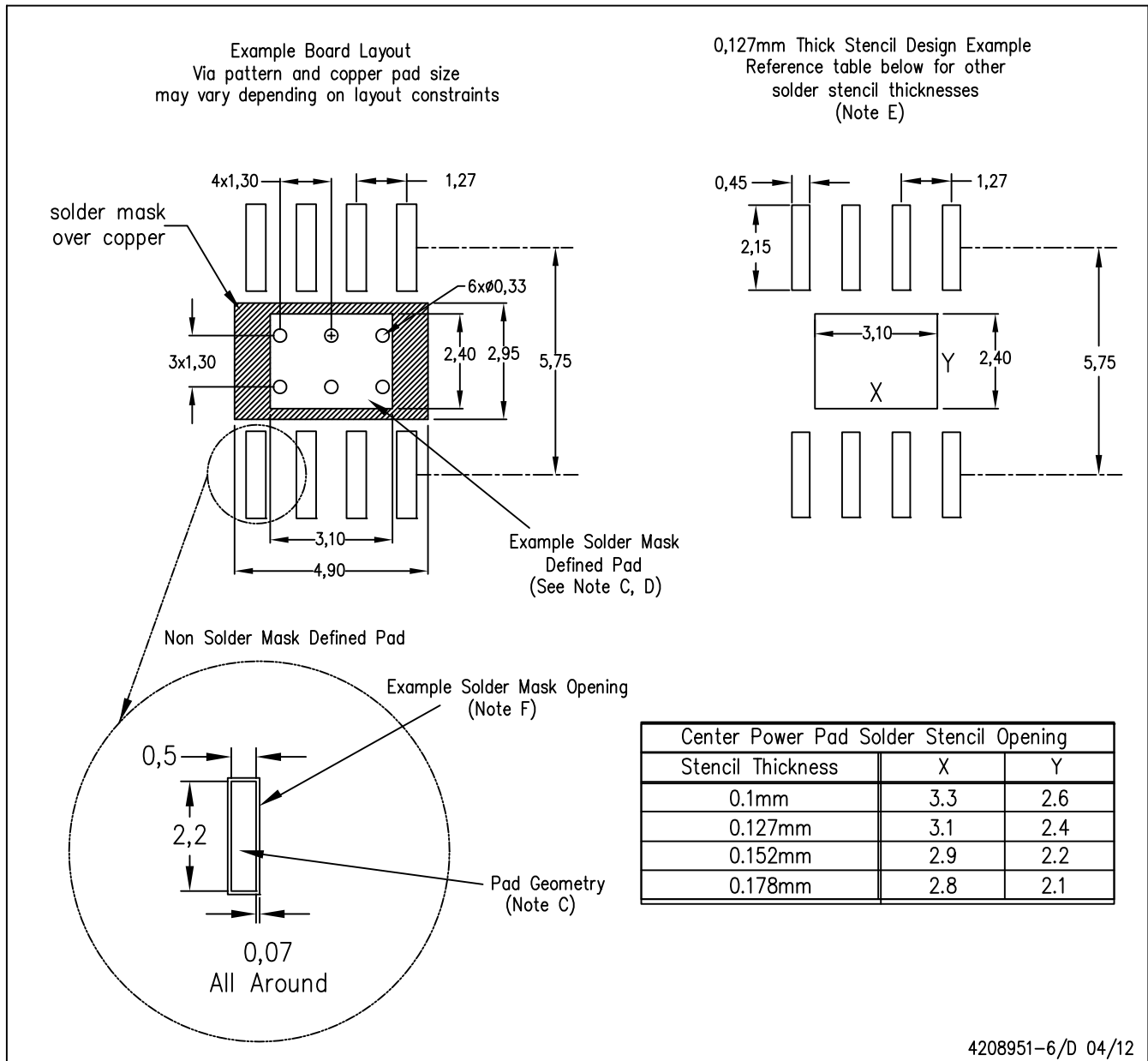
4206322-6/L 05/12

NOTE: A. All linear dimensions are in millimeters

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DDA (R-PDSO-G8)

PowerPAD™ PLASTIC SMALL OUTLINE



- NOTES:
- All linear dimensions are in millimeters.
 - This drawing is subject to change without notice.
 - Publication IPC-7351 is recommended for alternate designs.
 - This package is designed to be soldered to a thermal pad on the board. Refer to Technical Brief, PowerPAD Thermally Enhanced Package, Texas Instruments Literature No. SLMA002, SLMA004, and also the Product Data Sheets for specific thermal information, via requirements, and recommended board layout. These documents are available at www.ti.com <<http://www.ti.com>>. Publication IPC-7351 is recommended for alternate designs.
 - Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Example stencil design based on a 50% volumetric metal load solder paste. Refer to IPC-7525 for other stencil recommendations.
 - Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

PowerPAD is a trademark of Texas Instruments.

PowerPAD™ SOIC - 1.7 mm max height

Technical drawing of a microelectronic package showing top, side, and detail views with dimensions and callouts.

Top View:

- Overall width: 6.2 TYP, 5.8
- Overall height: 5.0, 4.8, NOTE 3
- Pin 1 ID Area: Indicated by a shaded square.
- Pin 1: Located at the top left.
- Pin 8: Located at the top right.
- Pin 5: Located at the bottom right.
- Pin 4: Located at the bottom left.
- Internal dimensions: 4.0, 3.8, NOTE 4
- Feature dimensions: 6X 1.27, 2X 3.81, 8X 0.51, 0.31
- Feature callout: $\oplus 0.25 \text{ M C A B}$

Side View:

- SEATING PLANE: Indicated at the top.
- Feature callout: 0.1 C
- Angle: 4X (0° - 10°)
- Width: 1.7 MAX

Detail A:

- Feature dimensions: 0.25, 0.10 TYP
- Callout: SEE DETAIL A

Bottom View:

- Overall width: 2.71, 2.11
- Overall height: 3.4, 2.8
- Pin 1: Located at the bottom left.
- Pin 8: Located at the bottom right.
- Pin 5: Located at the top right.
- Pin 4: Located at the top left.
- Exposed Thermal Pad: Indicated by a shaded square.

Detail A Typical:

- Feature dimension: 0.25
- GAGE PLANE: Indicated by a dashed line.
- Angle: 0° - 8°
- Feature dimensions: 1.27, 0.40
- Callout: DETAIL A TYPICAL

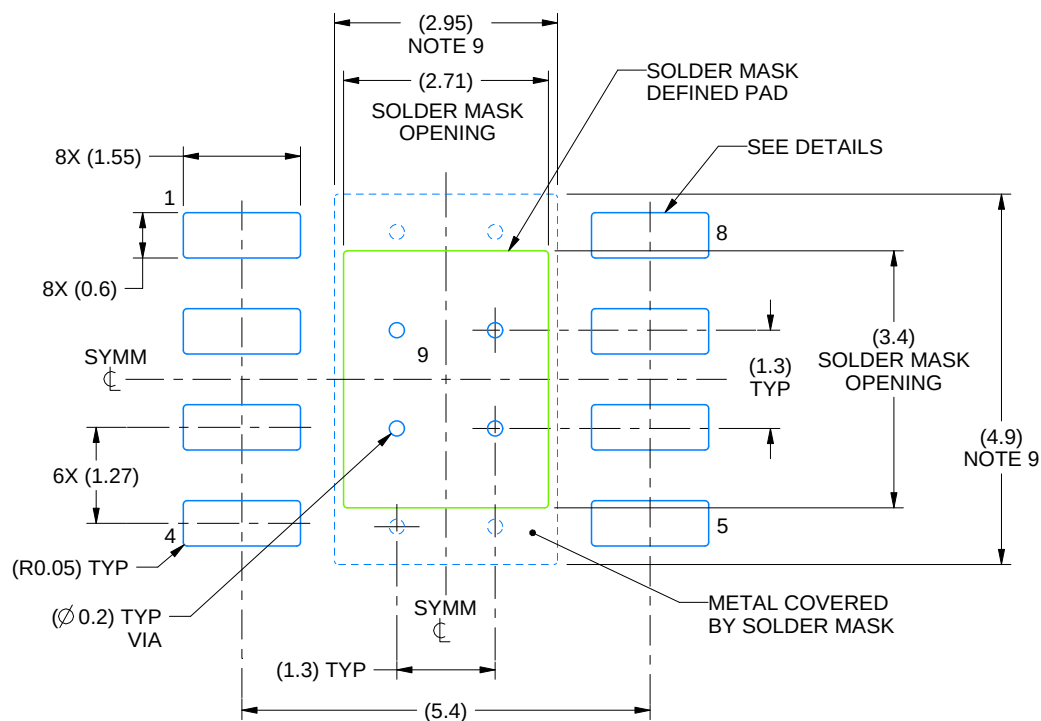
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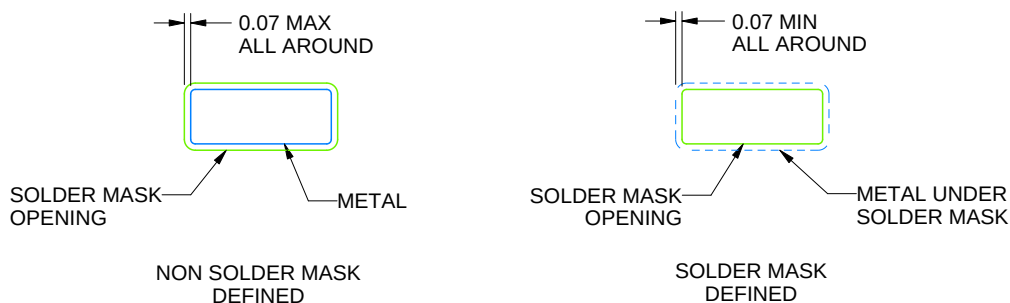
DDA0008B

PowerPAD™ SOIC - 1.7 mm max height

PLASTIC SMALL OUTLINE



LAND PATTERN EXAMPLE
SCALE:10X



SOLDER MASK DETAILS
PADS 1-8

4214849/B 09/2025

NOTES: (continued)

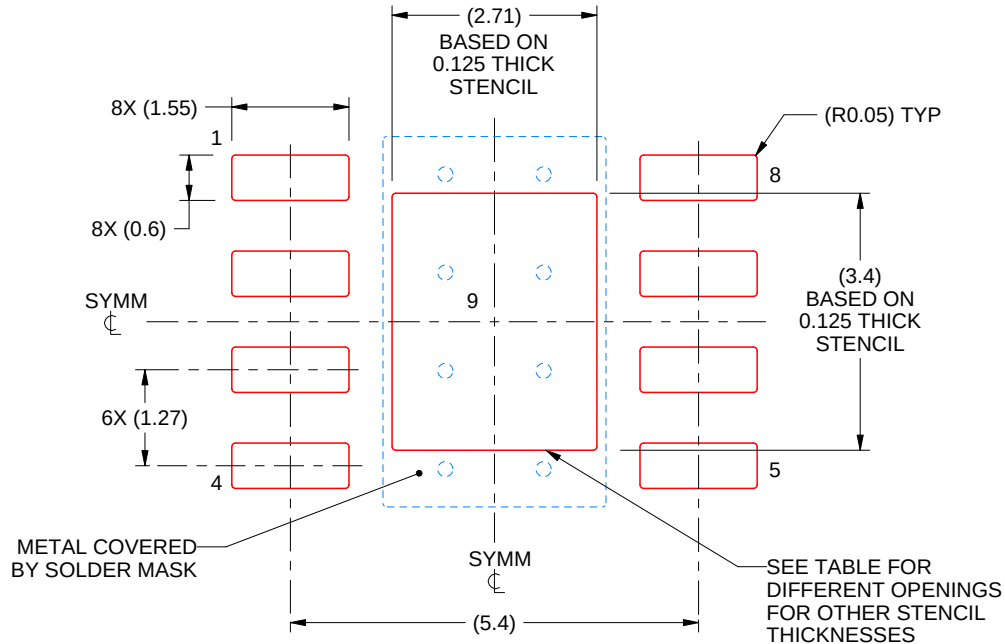
6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.
8. This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature numbers SLMA002 (www.ti.com/lit/slma002) and SLMA004 (www.ti.com/lit/slma004).
9. Size of metal pad may vary due to creepage requirement.
10. Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.

EXAMPLE STENCIL DESIGN

DDA0008B

PowerPAD™ SOIC - 1.7 mm max height

PLASTIC SMALL OUTLINE



SOLDER PASTE EXAMPLE
EXPOSED PAD
100% PRINTED SOLDER COVERAGE BY AREA
SCALE:10X

STENCIL THICKNESS	SOLDER STENCIL OPENING
0.1	3.03 X 3.80
0.125	2.71 X 3.40 (SHOWN)
0.150	2.47 X 3.10
0.175	2.29 X 2.87

4214849/B 09/2025

NOTES: (continued)

11. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
12. Board assembly site may have different recommendations for stencil design.

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最后更新日期：2025 年 10 月